

**Project Presentation of Industry Oriented Hands on Experience (22CS422)**

On

# FitTrack

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FitTrack is a web-based fitness and health management application designed to assist users in achieving their physical goals through intelligent workout scheduling and comprehensive routine tracking.

The system integrates personalized fitness planning with medicine reminder functionality, creating a unified platform that supports both physical wellness and health adherence.

Developed using React for the frontend and MongoDB for the backend, FitTrack offers a responsive, user-friendly interface combined with efficient data management.

The project demonstrates the application of modern web technologies in solving real-world health and fitness challenges, offering a scalable foundation for future enhancements such as AI-driven recommendations and wearable device integration.

Maintaining a healthy lifestyle requires consistent workout planning, routine tracking, and timely medicine intake — challenges many individuals struggle to manage effectively.

- Existing fitness applications focus solely on activity monitoring, while medication reminder apps operate separately — leading to fragmented health management.
- This lack of integration results in poor adherence to both fitness goals and medical schedules.
- Users need a unified system that simplifies workout scheduling, tracks daily progress, and ensures medication compliance.

FitTrack addresses this gap by combining fitness and medicine management into a single, user-friendly platform — promoting discipline, organization, and overall well-being.

# Objective of the Project

The objective of FitTrack is to provide a unified platform that:

- Generates personalized workout schedules tailored to individual goals such as weight loss, muscle gain, endurance improvement, or general fitness maintenance.
- Tracks daily fitness activities including completed workouts, calories burned, steps taken, and sleep patterns.
- Manages medicine reminders with automated alerts to ensure timely intake and treatment adherence.
- Enables users to maintain consistency, improve health discipline, and efficiently monitor progress toward their physical and wellness goals.

FitTrack aims to encourage healthier lifestyles, improve discipline, and support long-term wellness through a single integrated platform.

# Proposed Solution / System Overview



FitTrack is proposed as an integrated fitness and health management system delivering personalized workout scheduling and medicine tracking within a single platform.

The system collects user inputs — physical goals, fitness preferences, and medication details — and generates structured workout plans with timely reminders.

Major components of the system:

- User Management Module — secure registration, login, and profile management.
- Workout Scheduling Module — goal-based personalized exercise plan generation.
- Activity Tracking Dashboard — real-time monitoring of fitness metrics.
- Medicine Reminder Module — input medication details, dosage schedules, and alert timings.

The system also maintains secure data storage and user-specific records.

# Key Features

- **User Authentication & Profile Management:** Secure login, user profiles, and safe storage of personal fitness data.
- **Personalized Workout Scheduling:** Generates customized workout plans based on user-defined fitness goals for structured training.
- **Activity Tracking & Monitoring:** Tracks workouts, calories burned, steps taken, and overall routine consistency.
- **Progress Tracking & Dashboard:** Displays fitness progress through an interactive dashboard with visual insights and history management.
- **Medicine Tracker:** Allows users to input medication details and reminder timings, with browser notifications to ensure timely dose intake and streak tracking for consistency.

- **Frontend Development:** Built using React.js with responsive UI design and integration of charts (graphs and pie charts) for data visualization.
- **Backend Services:** Developed with Node.js and Express.js to handle server-side logic and RESTful APIs.
- **Database Management:** MongoDB is used for efficient and scalable data storage.
- **Authentication & Security:** Session-based authentication implemented using localStorage to maintain user login state.
- **API Integration:** RESTful APIs used for seamless communication between frontend and backend.
- **Data Visualization:** Libraries like Chart.js / Recharts used in React to represent fitness data through graphs and pie charts.

FitTrack is expected to deliver the following outcomes:

- Improved consistency in workouts and medication adherence through structured scheduling and reminders.
- Enhanced self-monitoring and progress visibility through an interactive dashboard.
- Promotion of healthier lifestyle habits and simplified routine management.
- Better organization and convenience for users managing both fitness and health conditions.
- Practical demonstration of integrating fitness and health management technologies into a scalable web application.

Users benefit from a streamlined, all-in-one platform that supports both physical and medical wellness.

FitTrack presents a novel approach by combining three key health management functions into a single unified system:

- Fitness Planning — personalized, goal-based workout scheduling.
- Routine Tracking — daily activity monitoring with visual progress insights.
- Medication Adherence — structured medicine reminders with streak tracking.

Unlike existing solutions that address only one of these areas, FitTrack integrates all three within a user-centric platform — creating a comprehensive health management experience accessible through a single web application.

This integration constitutes the original contribution of this project.

# Thank You